

From cohort to counterfactuals

How the CatBoost model is trained, tuned, evaluated and explained · 187 patients · random seed 42

■ used to fit ■ inner validation ■ outer test (held out) ■ search / decision

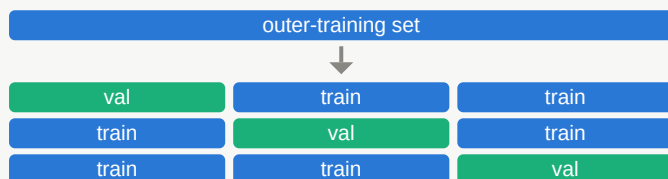
Which patients are held out

Repeated Stratified K-Fold: 4 folds × 10 repeats = 40 outer iterations



Which training patients tune the model

Stratified K-Fold: 3-fold, outer-training set only



Optuna TPE — 100 trials per outer iteration

Objective: mean AUPRC across the 3 validation folds

Tunes: depth · learning rate · L2 leaf regularization · positive-class weight

carried forward: winning hyperparameters + the same 3 inner splits

Refit, threshold selection, and outer-test evaluation

· per outer iteration

1 · Refit

Winning params refit on the full outer-training set

2 · Out-of-fold scores

One held-out score per patient

3 · Decision threshold

Maximize F1 on the OOF curve

4 · Outer-test evaluation

Refit model + its threshold scored once → performance metrics and plots

Explaining the model

· repeat 1 only: its 4 fold models and thresholds, saved and reused

Feature importance & SHAP

· the 4 fold models, refit

Refit with identical params, data and seed — reproduces the stored models

Feature Importance

get feature importance per fold model → 4-model heatmap

SHAP

Explainer over each fold's own test fold → mean |SHAP| heatmap

Counterfactuals (DiCE)

· the same 4 models, used as stored — no refit

Each wrapped with its own tuned threshold, so DiCE decides at that cut-off, not 0.5

Instances of interest

misclassified, or
 $|p - \text{threshold}| \leq 0.08$

DiCE genetic search

varies only the 6 actionable features

Outputs per fold

local CFs + global CF importance